



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AI Vibe Coding with C#

TGS Ref No: TGS-2023039178

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version v7.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
6.0	Date not stated in original PPT	Legacy content reference; original instructional topics retained.	Original source
7.0	13 September 2026	Rebuilt visual deck; corrected stale claims and wrong-course LOs; added ten C# activities, vibe prompts, tests, design and documentation depth. PPT/LG/LP/activities revised.	Dr Alfred Ang

TABLE OF CONTENTS

Course information.....	4
Learning outcomes and competency mapping.....	4
Teaching methods and evidence.....	4
Day 1 schedule.....	4
Day 2 schedule.....	4
Activity-to-slide reference.....	5
Assessment instruments and original contract.....	5
Facilitator preparation and review.....	5
Sources and change control.....	5

Course information

AI Vibe Coding with C# · TGS-2023039178 · ICT-DES-3005-1.1 Software Design Level 3. Two days / 16 contact hours = 7 classroom + 7 practical + 2 assessment. Lunch excluded; there is no separate break counted as instruction.

Learning outcomes and competency mapping

Outcome	Knowledge / ability
LO1: Determine basic software components using C# methodologies to meet functional specifications.	K1 / A1
LO2: Apply C# methodologies and tools for software creation.	K2 / A2
LO3: Select essential C# controls and features to meet software design requirements.	K3 / A3
LO4: Examine the interoperability and functionality of C# software components.	K4 / A4
LO5: Generate C# design documentation aligned with user specifications.	K5 / A5

Teaching methods and evidence

Each topic alternates a technical demonstration, didactic questions and independent activity. Review a concrete specification, code excerpt, design decision, failure path and test output. Activities are progressive and AI use is optional. Assess understanding by asking the learner to explain the invariant and reproduce a rejected path.

Day 1 schedule

Time	Block	Duration / slides
09:00–11:00	Topic 1: 60 min classroom + Activities 1–2 (30 min each)	120 min; slides 1–50
11:00–13:00	Topic 2: first 120 min of classroom/practical block	120 min; slides 51–137
13:00–14:00	Lunch	Excluded
14:00–15:00	Topic 2 completion: total classroom 90 min + Activities 3–4 (45 min each)	60 min; same Topic 2 range
15:00–18:00	Topic 3: classroom 90 min + Activities 5–6 (45 min each)	180 min; slides 138–176

Day 1 total: 480 contact minutes / 8 hours. Topics 1–3: 4 classroom hours + 4 practical hours.

Day 2 schedule

Time	Block	Duration / slides
09:00–12:00	Topic 4: classroom 90 min + Activities 7–8 (45 min each)	180 min; slides 177–213
12:00–13:00	Topic 5 first 60 min	60 min; slides 214–242
13:00–14:00	Lunch	Excluded
14:00–16:00	Topic 5 completion: total classroom 90 min + Activities 9–10 (45 min each); course wrap-up	120 min; same Topic 5 range plus worked charts/closing
16:00–17:00	WA-SAQ: 5 questions K1–K5	60 min
17:00–18:00	PP: 5 tasks A1–A5; LMS submission and summary sign-off	60 min

Day 2 total: 360 instructional minutes + 120 assessment minutes = 480 contact minutes / 8 hours.

Total programme = 840 instruction + 120 assessment = 960 minutes / 16 hours.

Activity-to-slide reference

Activity	Slide range	Outcome / evidence
01: From catalogue specification to executable contract	41–45	LO1 · K1 · A1; R1: blank names fail without creating a record.
02: Compile, run and review an AI-generated C# component	46–50	LO1 · K1 · A1; Whitespace and null input produce the learner fallback.
03: Types, operators, input rules and batch control flow	128–132	LO2 · K2 · A2; $19.90 \times 2 \times 0.90$ produces 35.82.
04: Methods and console MVC for balance enquiries	133–137	LO2 · K2 · A2; The Model rejects negative initial balance.
05: Catalogue classes, constructors and encapsulated properties	167–171	LO3 · K3 · A3; An invalid name, price, stock or category is rejected.
06: Interfaces, inheritance and polymorphic grade controls	172–176	LO3 · K3 · A3; All grades are within 0–100.
07: Order states, payment failure and notification guards	204–208	LO4 · K4 · A4; Success confirms and notifies exactly once in a single call.
08: JSON contracts, enum parsing, regex and file lifetime	209–213	LO4 · K4 · A4; A valid record survives JSON export/import unchanged.
09: Library methods, XML documentation and specification traceability	233–237	LO5 · K5 · A5; Book search is case-insensitive.
10: Documented stock component and handover evidence	238–242	LO5 · K5 · A5; $10 - 3 + 2$ produces stock 9.

Assessment instruments and original contract

Original papers retrieved from the verified course Drive Assessment folder because authenticated TMS edit-data has no assessment links. WA: exactly 5 questions, K1–K5 one-to-one, 60 min. PP: exactly 5 tasks, A1–A5 one-to-one, 60 min. Both are individual, open-book; all criteria must be demonstrated for competency. Each LU receives 12 min WA and 12 min PP in the approved proposal.

Facilitator preparation and review

Before class, verify SDK, all ten project builds and normal/invalid/boundary outputs. Check LMS resources and current candidate papers. Use the General Trainer card for the assigned facilitator; the named profile supplies the source developer identity. Capture actual activity evidence and provide formative feedback before assessment. Keep trainer answer keys outside the public learner tree.

Sources and change control

Source	Authority
Original PPT v6.0	reference/WSQ - Master Trainer Slides - Programming Methodologies in C# - v6.pptx
Approved proposal / TSC mapping	reference/extracted-original/CP_TIPL_Csharp_v1.1.txt
Registration page	https://www.tertiarycourses.com.sg/wsqaivibe-coding-with-c-sharp.html